

Physics:

2.2

1. Density:

mass per unit volume of substance is called density.

Formula / Equation:

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}} \quad \text{or} \quad \rho = \frac{m}{V}$$

It's SI unit stands as kilogram per cubic meter (Kg/m^3).

Smaller unit: gram per cubic centimeter (g/cm^3)

ii)

Volume of cuboid: $V = l \times w \times h$

Volume of cube: $V = a^3$

Sphere: $V = \frac{4}{3} \pi r^3$

iii)

The 2 cubes have same density because they are made of the same material, iron. The large cube has greater mass and volume in the same proportion. So mass per unit volume remains unchanged.

Q:3
i)

$$\text{mass of liquid} = \text{total mass} - \text{mass of cylinder}$$
$$160 - 120 = 40\text{g}$$

ii) Density of liquid :-
formula: $\rho = \frac{m}{V}$

$$\rho = \frac{40}{50} \Rightarrow 0.8 \text{ g/ml}$$

iii) convert to Kg/m^3

$$1 \text{ g/ml} \Rightarrow 1000 \text{ Kg/m}^3$$
$$0.8 \times 1000 = 800 \text{ Kg/m}^3$$

full sentence:-

The density of liquid is 800 Kg/m^3

Q:4

i)

	Solid	Liquid	Gas
Arrangement	Closely packed, tightly packed.	Packed but Passing.	flowing
movement	It is in near position but vibrate.	Their position change.	Random motion.
Density	High	medium (intermediate)	Low

The attractive forces between particles in a solid are strong, so the particles are held very close together. In a gas, the attractive forces are very weak, so the particles are far apart.

Therefore, a solid has much more mass in the same volume and hence has a much higher density than a gas.

Q.5

i) Initial water level = 30 ml

Final water level = 55 ml

$$V = 55 - 30 \\ = 25 \text{ ml}$$

$$\text{Since } 1 \text{ ml} = 1 \text{ cm}^3 \\ = 25 \text{ cm}^3$$

$$\text{ii) } \rho = \frac{m}{V}$$

$$= \frac{175}{25} = 7 \text{ g/cm}^3$$

iii) method :-

The method is the water displacement method.

* It is used for an irregularly shaped solid.

* The object must sink completely in water.

* It must not dissolve / react with water.

$$\begin{aligned} 20\text{cm} &= 0.20\text{m} \\ 10\text{cm} &= 0.10\text{m} \\ 5\text{cm} &= 0.05\text{m} \end{aligned}$$

Volume :-

$$V = lwh$$

$$= 0.20 \times 0.10 \times 0.05$$

$$0.001\text{m}^3$$

i) Density :-

$$\text{mass} = 19.3\text{kg}$$

$$\rho = \frac{m}{V}$$

$$= \frac{19.3}{0.001}$$

$$= 19300 \text{ kg/m}^3$$

vi) metal :-

The metal is tungsten, cuz
it's density is approxiam-
etdy 19300 kg/m^3 .

full sentence is tungsten.
The metal is tungsten.

b. Efficiency

i) Definition and formula:

Efficiency is the ratio of useful output energy to total input energy, usually expressed as Percentage.

$$\text{Percentage Efficiency} = \frac{\text{Useful output energy}}{\text{input energy}} \times 100$$

ii)

Input energy : 2500 J

Output energy : 1500 J

$$\text{Efficiency} = \frac{1500}{2500} \times 100$$

$$= 60\%$$

iii) Useful output Power:-

$$P_{\text{out}} = \eta \times P_{\text{in}}$$

Efficiency as a decimal:-

$$\eta = 60\% = 0.60$$

$$P_{\text{out}} = 0.60 \times 600$$

$$P_{\text{out}} = 360 \text{ W}$$

iv) A machine cannot have 100%.

but some of the input energy is always wanted, for example as heat or sound, due to friction and other energy losses.

and the name of device which have lowest efficiency:-

Device	⇒	Efficiency
Incandescent light	⇒	5%

MCQs:-

1. C
2. A
3. B
4. D
5. C
6. B
7. A
8. ~~B~~ D
9. C
10. B